

Lab 06: Design of a McPherson suspension

31/10/2025

Abstract

This report concerns with the design and structural validation of a McPherson suspension system. The process begins with a kinematic optimization to determine the coordinates of characteristic points that meet target camber and steering angle requirements.

The lower control arm is designed through a multi-stage approach: preliminary load estimation under extreme conditions, topological optimization to minimize mass, and final validation via FEM analysis. The report also includes detailed production specifications, including tolerances and subsystem assembly drawings.

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1. Introduction

This technical report covers the development of a McPherson suspension system, following a workflow that integrates theoretical simple modelling with computational validation and optimization.

The project is structured into three main phases:

- Kinematic Synthesis: a neural network and MATLAB's `fmincon` function are used to optimize the spatial positioning of characteristic points to achieve target camber and steering angle behaviors
- Structural Design of the Lower Arm: estimation of reaction forces at the chassis and hub under high operating loads, such as longitudinal braking forces and vertical dynamic bumps
- Optimization and Validation: a topological optimization is used to minimize material volume by at least 70% while maximizing stiffness, followed by a final FEM analysis to validate the stresses on the forged Al 6061 component.

Beyond the control arm, the report provides a comprehensive overview of the full assembly, including the design of the cast wheel rim and the mechanical interfaces for the damper, hub carrier, and constant velocity joint.

2. Positioning of the characteristic points

The first step in the design of a suspension system is the positioning of the characteristic points that define the outlines of the configuration; these points are highlighted in Figure 1 with letters A-G (8 points + point O that represents the origin of the reference system).

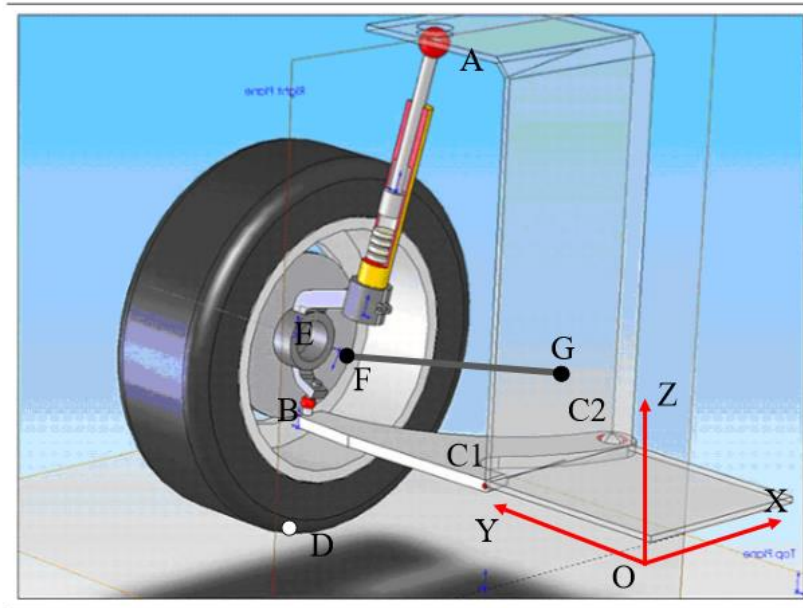


Figure 1

The coordinates of some points are given by the assignment while others need to be tuned in order to respect some target values of camber and steer angle when the suspension is in full compression or full extension; this is performed through an optimization process:

- Design variables:
 - $C1_y \rightarrow y$ coordinate of joint 1 between chassis and lower arm
 - $C1_z \rightarrow z$ coordinate of joint 1 between chassis and lower arm
 - $G_y \rightarrow y$ coordinate of the joint between the steering rod and the chassis
 - $B_y \rightarrow y$ coordinate of the joint between the hub carrier and the lower arm
 - $B_z \rightarrow z$ coordinate of the joint between the hub carrier and the lower arm
- Objective function to be minimized:
 - Sum of the mean squared errors between the imposed values of camber and steering angle and the values computed as a function of the design variables
- Upper and lower bounds of the design variables are provided by the assignment
- No constraints on the objective function are present
- In the optimization process, a neural network is used to replicate the behavior of the suspension when the coordinates of these characteristic points are varied (it replicates

the non-linear relationship between the coordinates of the points and the camber and steering angle without having to run a multibody simulation at each iteration)

- *fmincon* MATLAB function is used with the following settings:
 - 'Algorithm','sqp',
 - 'TolX', 1e-9

A first minimization is performed and the results are reported in Figure 2. Note that the result changes if the initial guess is changed; this indicates that a relative minimum is found (not the absolute minimum). In reality, this is exploited to have more control over the obtained solution: in case the current design violates some volume/clearance constraints that have been given, modifying the initial condition accordingly allows to find another solution that suits the requirements.

Case 1	Punti	X [m]	Y [m]	Z [m]
	A	31.80	593.00	604.00
	B	-9.10	716.05	-55.70
	C1	-10.00	399.10	-49.60
	C2	311.00	399.10	-45.00
	D	-1.60	773.50	-226.80
	E	-1.60	768.40	65.10
	F	136.90	682.70	88.50
	G	170.00	320.00	86.00
	O	0.00	0.00	0.00

Scrub Radius [mm]	25.00
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Figure 2

For example, the first design implies having a scrub radius that is $\approx 25mm$; since it was suggested to try and achieve a lower value, the initial guess is changed to obtain a higher value of B_y and a lower value (in absolute value) of B_z . The optimization process is repeated and the results obtained are reported in Figure 3.

Case 2	Punti	X	Y	Z
	A	31.80	593.00	604.00
	B	-9.10	727.85	-53.97
	C1	-10.00	408.76	-48.01
	C2	311.00	408.76	-43.41
	D	-1.60	773.50	-226.80
	E	-1.60	768.40	65.10
	F	136.90	682.70	88.50
	G	170.00	319.50	86.00
	O	0.00	0.00	0.00

Scrub Radius [mm]	10.23
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Figure 3

Figure 4 proofs that the optimization process is successful: the camber and steering angle curves as a function of the wheel vertical travel pass through the red dots representing the target values.

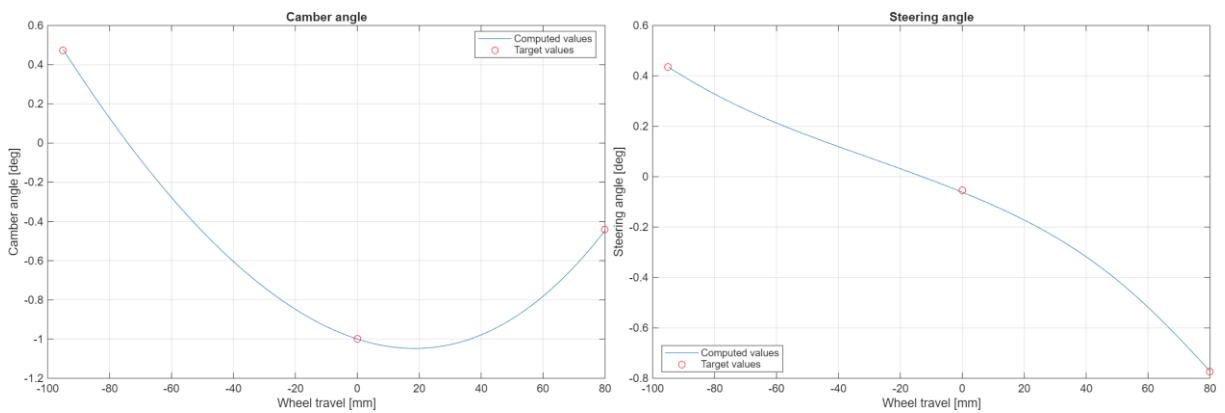


Figure 4

3. Design of the lower arm

The design of the lower arm is carried out in two steps: first a rough estimation of the forces exchanged and stresses is performed; successively, the result is refined through a topological optimization and FEM analysis.

3.1 Computation of the loads acting on the lower arm and the reaction forces exchanged with the chassis

In this preliminary analysis, the lower arm is modelled as a set of beams connected between each other by means of hinges.

The steps required to compute the reaction forces exchanged with the chassis are the following

- Estimation of the forces exchanged between the tire and the contact patch (Section 3.1.1)
- Computation of the forces exchanged in points A and (more interestingly) point B (Section 3.1.2)
- Computation of the reaction forces exchanged in points C1 and C2 (Section 3.1.3)

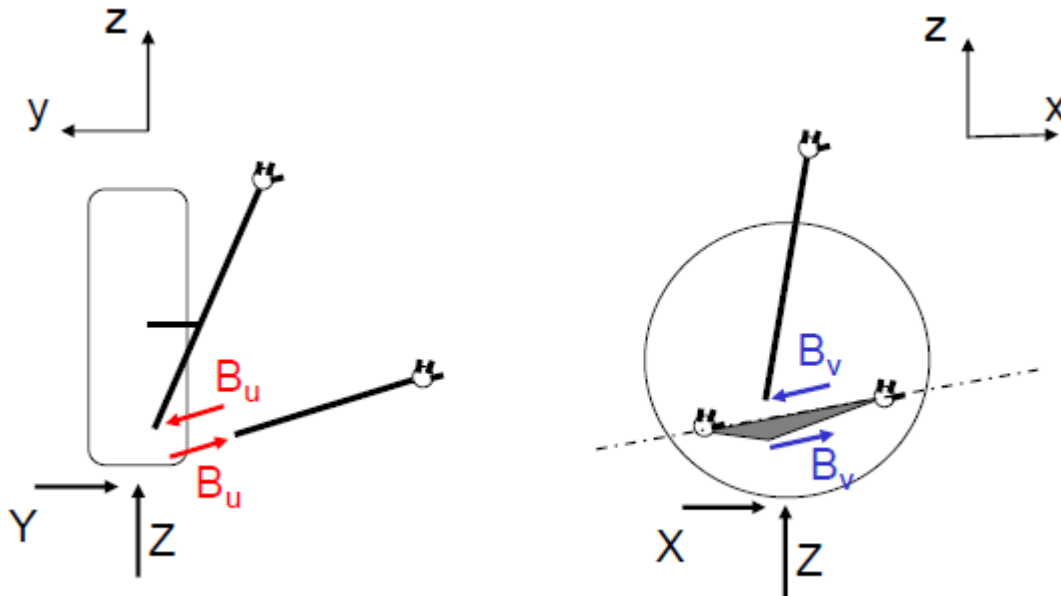


Figure 5

3.1.1 Contact patch forces for the most demanding situation

- Vertical force

$$Z = k \cdot F_{z,stat}$$

- k takes into account dynamic effects due to road unevenness or bumps (2.5 for the safety of the suspension)
- Lateral force
 - $Y = \mu_y \cdot Z$ ($\mu_y = 0.7$)
 - This condition refers to the vehicle hitting a bump when it is turning with high lateral acceleration
- Longitudinal force
 - $X = \mu_x \cdot Z$
 - This condition considers traction and braking

3.1.2 Forces exchanged in point A, B

Forces and moment equilibrium for the B-A bar in the y-z and x-z plane are considered (Figure Figure 6) . The angles as well as the beam lengths are defined starting from the coordinates of the points computed in Section 2.

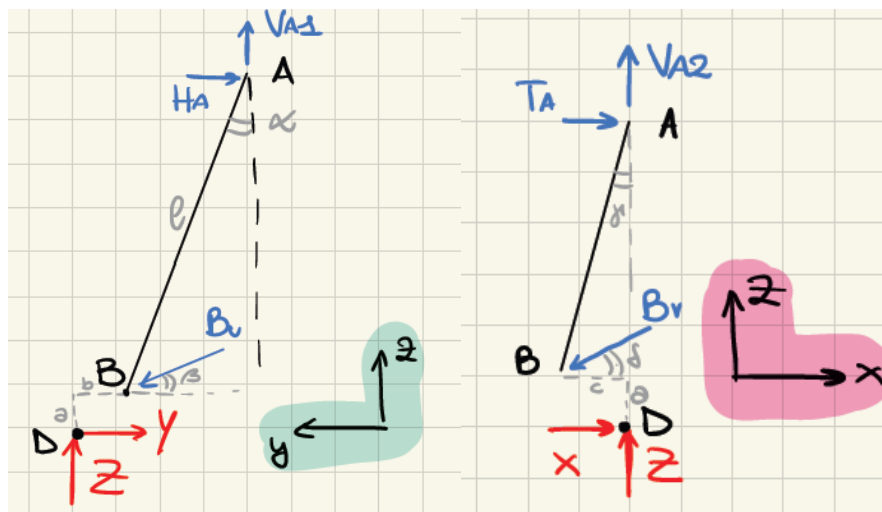


Figure 6

Writing the forces and moments balance equations leads to the resolution of the following system, where the unknowns are highlighted in yellow:

$$\left\{ \begin{array}{l} \sum F_{y_{yz}} \rightarrow Y + H_A - B_U \cos(\beta) = 0 \\ \sum F_{z_{yz}} \rightarrow Z + V_{A_{yz}} - B_U \sin(\beta) = 0 \\ \sum M_{B_{yz}} \rightarrow Yb - Za - H_A l \cos(\alpha) + V_{A_{yz}} l \sin(\alpha) = 0 \\ \sum F_{x_{xz}} \rightarrow T_A + X - B_V \cos(\delta) = 0 \\ \sum F_{y_{xz}} \rightarrow V_{A_{xz}} + Z - B_V \sin(\delta) = 0 \\ \sum M_{B_{xz}} \rightarrow Xa - Zc - T_A l \cos(\gamma) + V_{A_{xz}} l \sin(\gamma) = 0 \\ V_A = V_{A_{yz}} + V_{A_{xz}} \end{array} \right.$$

The computed values of the reaction forces in B are reported below:

B_u	4271.6 [N]
B_v	13966 [N]

3.1.3 Forces exchanged in points C1, C2

Forces and moment equilibrium for the lower arm in the x-y plane are considered (Figure 7):

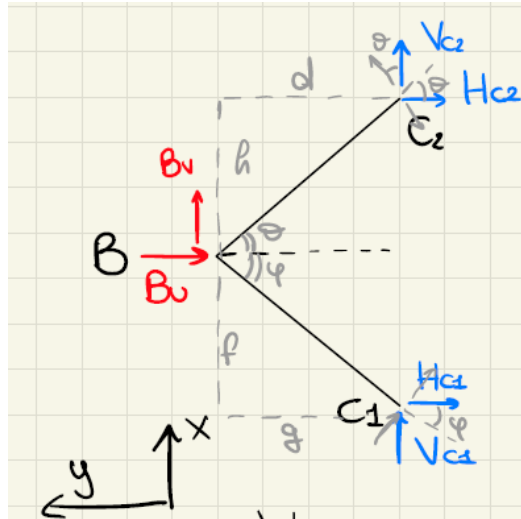


Figure 7

Writing the forces and moments balance equations leads to the resolution of the following system, where the unknowns are highlighted in yellow:

$$\left\{ \begin{array}{l} \sum F_{x_{yx}} \rightarrow V_{c1} + V_{c2} + B_v = 0 \\ \sum F_{y_{yx}} \rightarrow B_u + H_{c2} + H_{c1} = 0 \\ \sum M_{B_{yx}} \rightarrow V_{c2} * d - H_{c2} * h + H_{c1} * f + V_{c1} * g = 0 \\ V_{c2} = K_{c2} * v \\ V_{c1} = K_{c1} * v \\ K_{c1} = K_{c2} \end{array} \right.$$

The ratio between K_{c1} and K_{c2} drives the design of the bushings connecting the lower arm to the chassis. A typical bushing has a radial stiffness that is much higher than the axial stiffness. If the selected ratio between the stiffnesses is high, it means that their axes are orthogonal. If the selected ratio between the stiffnesses is close to 1, it means that they are coaxial; [this is the choice that has been made](#). The computed values of the reaction forces in C1, C2 are reported in the table below:

$H_{c1,xy}$	9622.9 [N]
$V_{c1,xy}$	-6982.9 [N]
$H_{c2,xy}$	-13894 [N]
$V_{c2,xy}$	-6982.9 [N]

3.2 Selection of manufacturing process and material

The lower arm is going to be manufactured by forging and the material is going to be an aluminum alloy with the following characteristics:

- Al 6061 alloy
 - Density: $2.7 \cdot 10^{-9} \text{ ton/mm}^3$
 - Elastic properties
 - $E = 70'000 \text{ MPa}$
 - Poisson ratio 0.3
 - Yield strength $\sigma_y = 240 \text{ MPa}$
 - Heat treated and naturally aged
 - Magnesium and Silicon as alloy elements
 - Corrosion resistance
 - Good strength and weldability

3.3 Internal forces, stresses and minimum cross section dimensions

After having selected the material and the manufacturing process (hence the shape of the cross section), it is important to estimate the stresses and the minimum cross section dimensions; to do this, the x,y plane is considered (top view of the lower arm).

Note that the vertical rod connecting C1, C2 is not considered because C1, C2 are connected to the vehicle chassis and not directly between each other. Figure 8 and Figure 9 display the computation of the internal forces (axial, shear and bending moment)

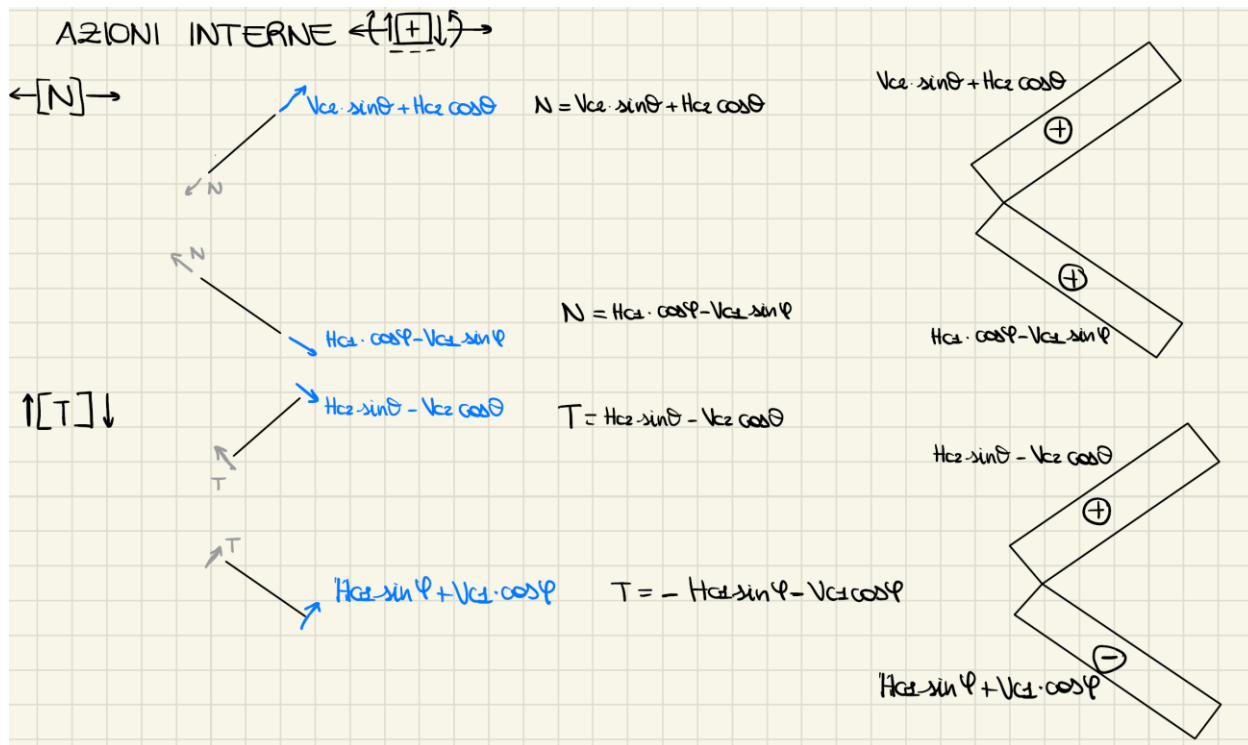


Figure 8

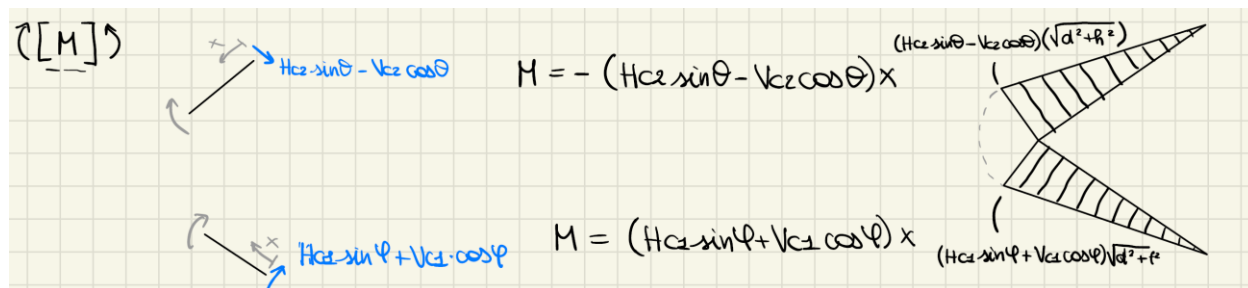


Figure 9

The most stressed point is point B; here the internal forces are (pedix C1 indicates it is computed along the beam that connects B with C1; pedix C2 indicates it is computed along the beam that connects B with C2):

- $N_{C1} = H_{C1}\cos\varphi - V_{C1}\sin\varphi = 9642.5 \text{ N}$
- $N_{C2} = V_{C2}\sin\theta + H_{C2}\cos\theta = -14755 \text{ N}$
- $M_{C1} = M_{C2} = M = (H_{C1}\sin\varphi + V_{C1}\cos\varphi) * \sqrt{d^2 + f^2} = (H_{C2}\sin\theta - V_{C2}\cos\theta) * \sqrt{d^2 + h^2} = -2219.5 \text{ N}$

The yielding stress of selected material is $\sigma_Y(Al\ 6061) = 240 \text{ MPa}$. To compute the minimum cross section dimensions, the worst possible case is considered $\rightarrow$ the two axial loads are summed and a safety factor of 1.5 is introduced (SF = 1.5).

As a first approximation the cross section is assumed to be squared and the following result is obtained:

$$\sigma = \frac{N_1}{lato^2} + \frac{N_2}{lato^2} + \frac{M}{\frac{1}{12}lato^4} * \frac{lato}{2} \leq \frac{\sigma_{adm}}{\text{safety factor}} \Rightarrow \text{lato} \geq 21.7 \text{ mm}$$

To get a more relevant and usable result the manufacturing process is taken into account: since the lower arm is going to be forged, the cross section is going to be roughly H shaped. An optimization process is used to tune the cross section dimensions:

- Design variables: H shaped beam dimensions reported in Figure 10
 - o $h_s \rightarrow$ beam height
 - o $b_s \rightarrow$ flange width
 - o $t_{fs} \rightarrow$ flange thickness
 - o $t_{ws} \rightarrow$ web thickness

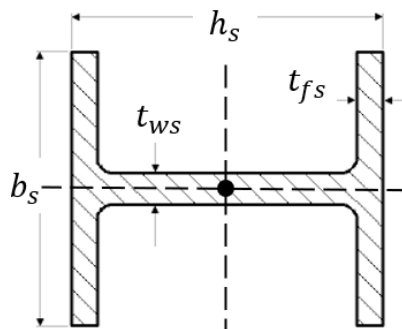


Figure 10

- Objective function to be minimized:
 - o Cross section area
- Lower and upper bounds:

- Lower bounds:
 - $l_b = [0.0200, 0.0200, 0.0060, 0.0060] [m]$
 - some of them are set $\neq 0$ to account for the fact that the workpiece is going to be forged and it cannot be too thin
- Upper bounds:
 - $u_b = [0.08, 0.08, 0.02, 0.05] [m]$
- Constraints:
 - Stress constraint: $\sigma \leq \frac{\sigma_{adm}}{safety\ factor} \rightarrow \frac{N_1}{A(x)} + \frac{N_2}{A(x)} + \frac{M}{J_z(x)} * \frac{h_s}{2} - \frac{\sigma_{adm}}{safety\ factor} \leq 0$ (z is in the suspension reference system, the one taken on the website is J_x)
 - Geometry constraint: $A(x) > 0 \rightarrow -A(x) < 0$
 - Geometry constraint: $h_s - 2t_{fs} > 0 \rightarrow 2t_{fs} - h_s < 0$
 - Geometry constraint: $b_s - t_{ws} > 0 \rightarrow t_{ws} - b_s < 0$
- Initial guess:
 - x_0 selected as average between lower and upper bound to make sure it respects the constraints

The results of the optimization are the following:

h_s	80 [mm]
b_s	25.7 [mm]
t_{fs}	6 [mm]
t_{ws}	6 [mm]

Note that these results represent a rough estimation based on a high overestimation of the loads, the real cross section dimensions in point B might not be precisely these ones but the order of magnitude/the proportions are still respected.

3.4 Topological Optimization

The aim of the topological optimization is to find the optimal distribution of material inside the component. The steps carried out are the summarized in the following points

- 1) Import elementary cad of lower arm (Figure 11) → the cad model must have more material than the final one
 - The optimization process is not going to cancel nodes/points; it is assigning to each point a relative density. High relative density is assigned where material is needed

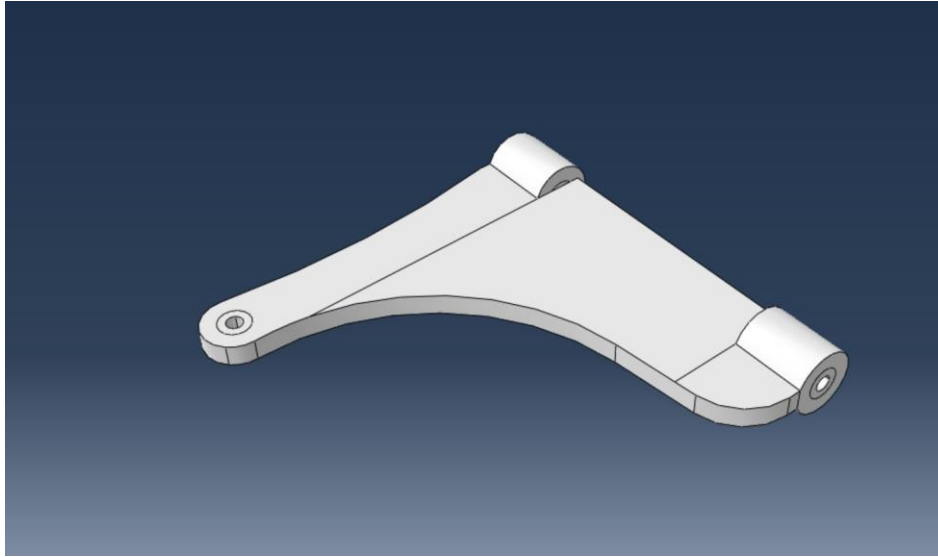


Figure 11

- 2) Assign the material (Al)
- 3) Create the points where the forces and the boundary conditions are applied
 - 2 points are the bushing where lower arm is connected to the chassis → assigning a property to those points is equal to assigning a property to the internal surface of the two bushings housing. The boundary condition is going to be assigned to the two points just defined
 - Third point is the connection between the lower arm and the hub carrier (point B, where the forces B_u , B_v will be applied) → assigning a property to that point is equal to assigning a property to the curved surface around point B. B_u , B_v will be applied in that point
- 4) Create the mesh
 - Approximate global size: 4
 - Whole mesh tetrahedral (no concentrated refinement is applied)
- 5) Setup the topological optimization
 - Freeze load regions
 - Where loads are applied material is not removed by the optimization
 - Two parameters need to be defined (one is going to become a constraint and the other one is going to be the objective function)
 - Strain energy → objective function to be minimized = maximize stiffness
 - Volume → constraint (it needs to decrease at least by 70%)
 - Geometrical constraint
 - Forging

The results of the optimization are reported in Figure 12, Figure 13

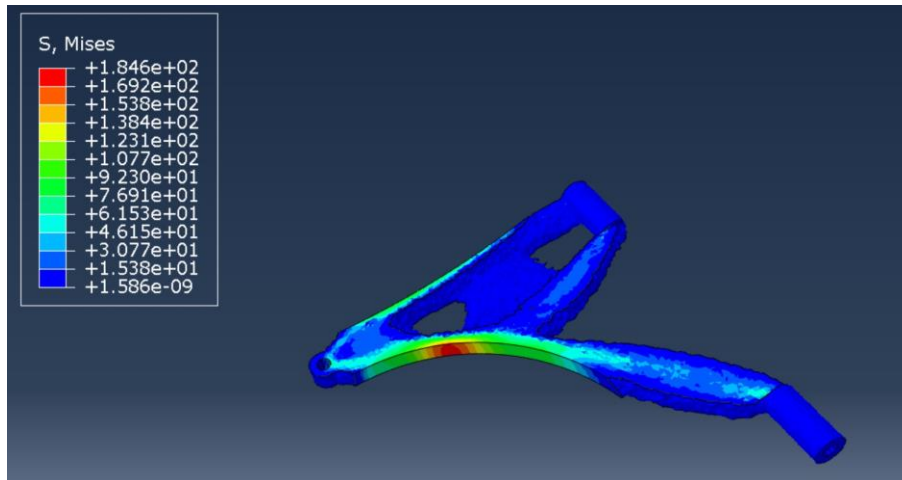


Figure 12

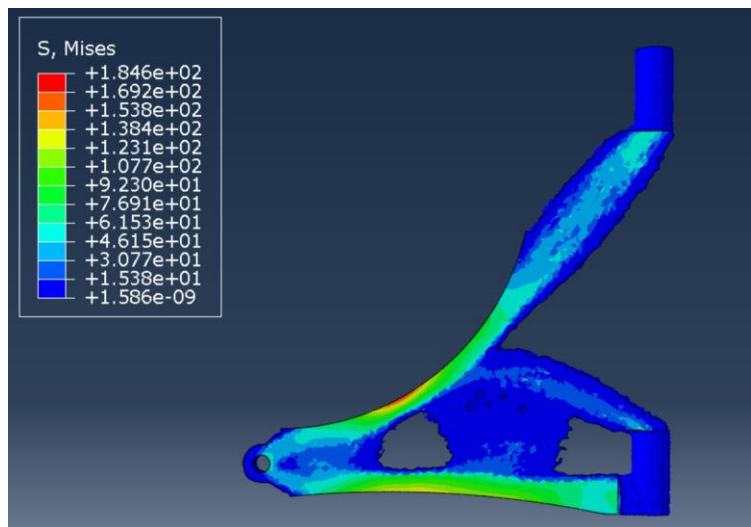


Figure 13

3.5 FEM analysis

The result of the topological optimization is taken as an indication of the shape the component needs to have. The quality of the final design is validated by means of a FEM analysis, that is performed on the design reported in Figure 14.

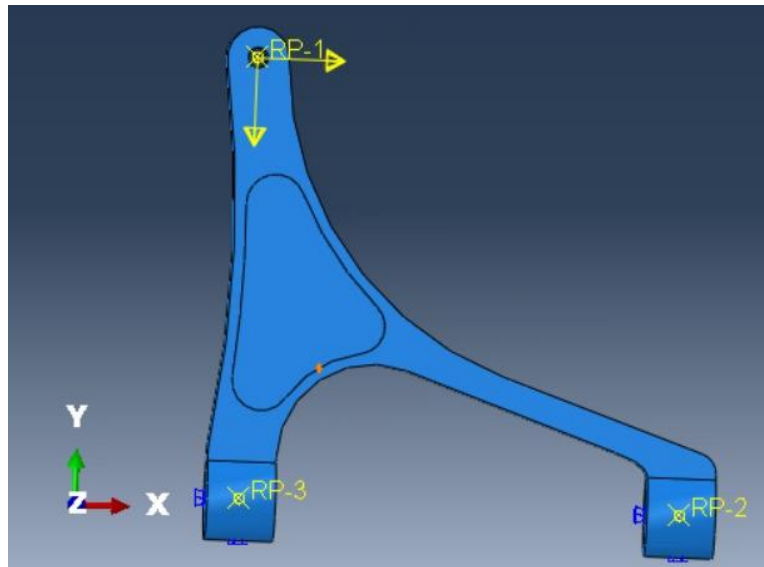


Figure 14

The steps carried out are summarized in the following points:

- 1) Import cad of final design of the lower arm
- 2) Assign the material (Al)
- 3) Step
 - Static: means that inertia effects are considered negligible
 - General : means the software can handle non linearities.
- 4) Constraints
 - In point B
 - o Encaster
 - In point C2
 - o Encaster (in plane analysis is considered)
 - In point C1
 - o Encaster (in plane analysis is considered)
- 5) Loads
 - in point B $\rightarrow B_u, B_v$ are applied
 - no loads in C1, C2 because the reaction forces are internally computed by the solver
- 6) Mesh (Figure 15)
 - Approximate global size: 2.5 mm
 - Tetraedric mesh, quadratic shape function

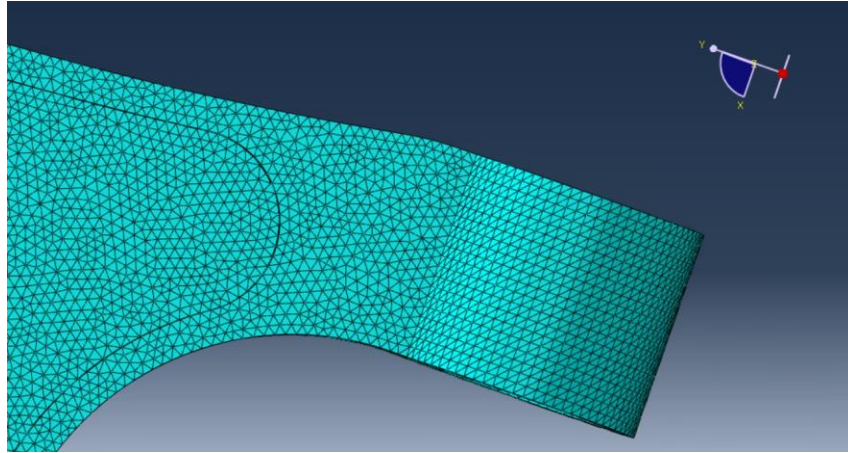


Figure 15

7) Results

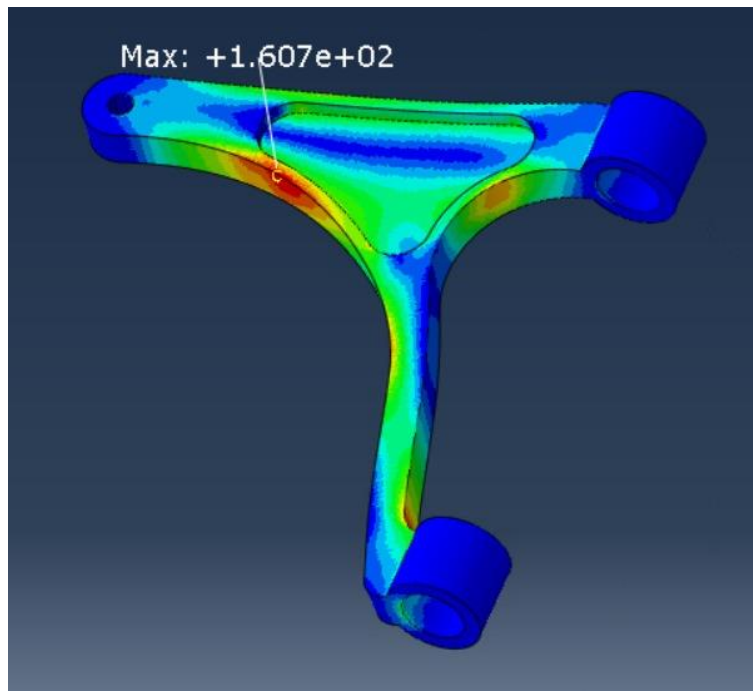


Figure 16

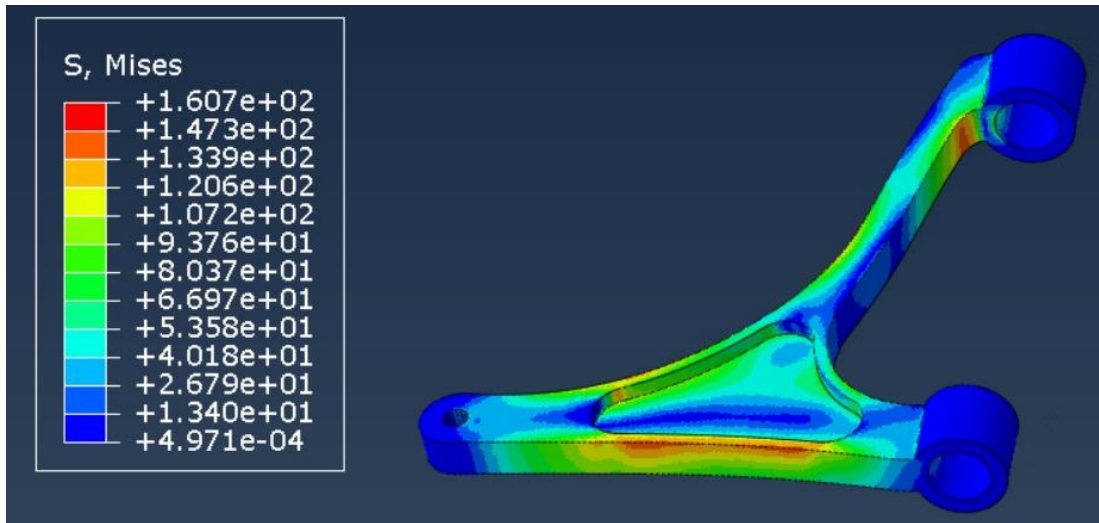


Figure 17

The maximum vertical deflection is in point B is $\approx 2.3 \text{ mm}$

4. Production drawing of the lower arm

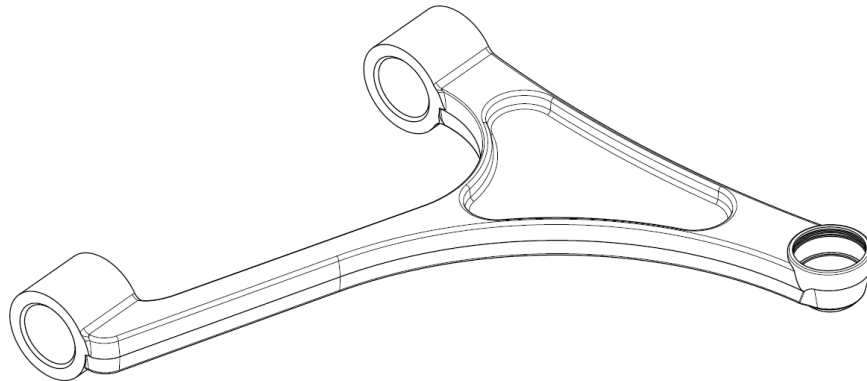


Figure 18

4.1 Details of lower arm

4.1.1 Tolerances

The component is going to be forged and successively machined to achieve the required tolerances

- Surface roughness
 - $0.8 \mu\text{m}$ for the inside surface of the bushings connecting the lower control arm with the chassis. Material removal process is used (source: SKF website).
- Geometric tolerances (reference to section A-A):
 - **Cylindricity** Ⓢ : it requires the entire machined surface of the hole to lie between two concentric cylinders spaced 0.05 mm apart. It limits ovality, taper, and straightness errors simultaneously
 - **Planarity** Ⓜ : it creates a tolerance zone consisting of two parallel planes spaced 0.1 mm apart. The face it refers to must lie inside this 0.1 mm wide zone
 - **Concentricity** Ⓢ : it controls the alignment of the second hole relative to the first (Datum A). It requires the axis of the right bore to lie within a cylindrical tolerance zone of 0.05 mm diameter centered on the axis of the left hole
 - **Perpendicularity** ⊥ : it requires the pointed face to be perpendicular to the hole axis (Datum A) within a tolerance zone of 0.1 mm (the surface it is applied to must be within two planes 0.1 mm apart orthogonal to Datum A)
- Dimensional tolerances
 - High precision grade is used for the bushing holes: 40H6 implies that the dimension of the hole must be $40 \text{ mm} \div 40.016 \text{ mm}$ (hole dimension cannot be smaller than nominal)

- High precision is required in overall lower arm width and in the width of the mounting bosses ($\pm 0.1 \text{ mm}$); the reason for this is that the component must fit between two fixed brackets on the vehicle chassis
- Looser tolerances are given to locate the position of the spherical joint in the top view ($\pm 0.2 \text{ mm}$). While the position of the ball joint (wheel hub connection) is important for suspension kinematics (camber/caster), it connects to a movable steering knuckle rather than a fixed rigid bracket, allowing for slightly less strict tolerance

4.1.2 Connection between lower arm and hub carrier

The lower arm is “hanging” through the spherical joint from the hub carrier. The weight of the car is transferred from the chassis → spring + damper → hub carrier → hub → wheel; the lower arm does not support the weight of the car (as it would happen in a double wishbone). This is the reason for the adopted design choice.

The housing for the spherical joint in the lower arm has plastically deformed tips that hold in position the polymeric material and the ball joint; a rubber cover that prevents dust and dirt from entering the housing is also present.

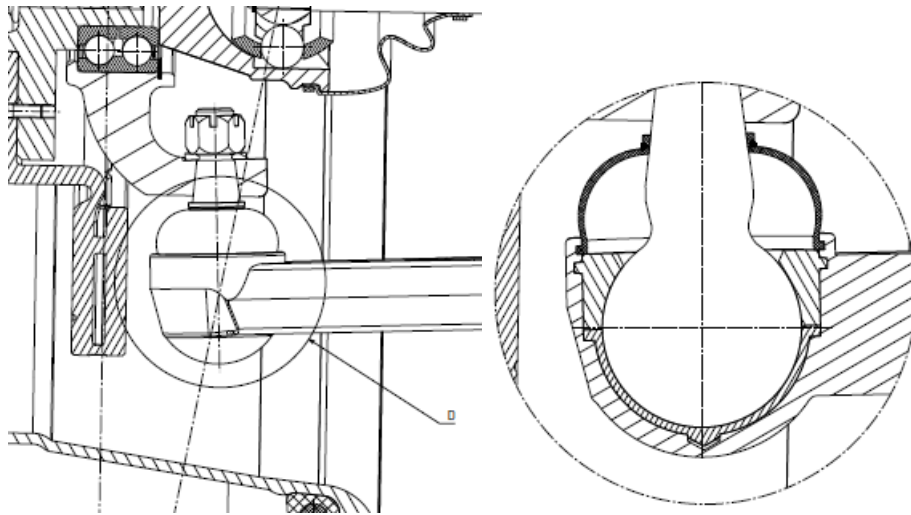


Figure 19

4.1.3 Connection between lower arm and chassis

Coherently with the model used in Section 3.1.3, the two bushings have parallel axes. The structure of the bushings is reported in Figure 20:

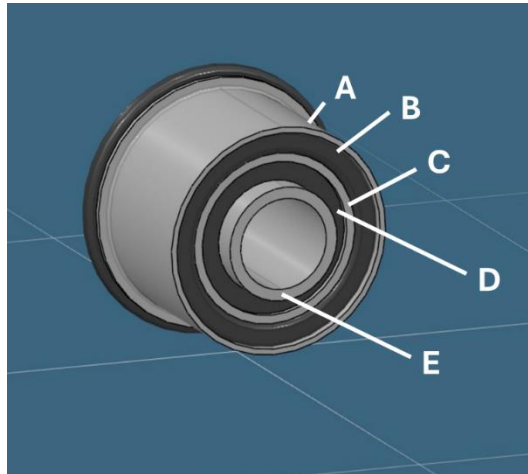


Figure 20

- Outer metal sleeve that is press fitted in the hole on the lower arm **(A)**
- Polymeric material **(B)**
- Metal sleeve **(C)**
- Polymeric material **(D)**
- Inner sleeve **(E)** → the through mounting bolt passes through this sleeve

To prevent axial displacement of the lower arm, one tip of the outer sleeve forms a flange. This feature ensures the bushing cannot be pushed through the arm and provides the necessary surface area to transfer axial loads from the arm to the vehicle chassis (Figure 20, Figure 21).

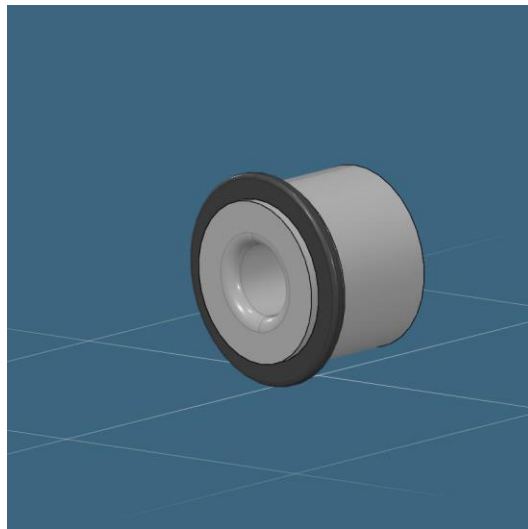


Figure 21

5. Overall full-scale drawing of the suspension

5.1 Details of the whole suspension

5.1.1 Wheel rim

The tire dimensions are given by the assignment: 205/60 R15; based on the provided “*tire size and rim size correlation table*” (partially reported in Figure 22), the rim needs to have width between $7\text{ in} \div 7.5\text{ in}$ ($7J, 7J\frac{1}{2}$ rim). The designed rim has width $7\text{ in} = 178\text{ mm}$.

7.0 inches	195 mm	205 or 215 mm	225 mm
7.5 inches	205 mm	215 or 225 mm	235 mm

Figure 22

The other rim characteristics are summarized in the following points:

- Rim diameter is constrained by the adopted tire model (15 in)
- Rim outset: 40 mm
- The hump that is needed to prevent bead unseating (essential for vehicle safety) is designed according to the directions given by the standard tables (Figure 23, Figure 24)

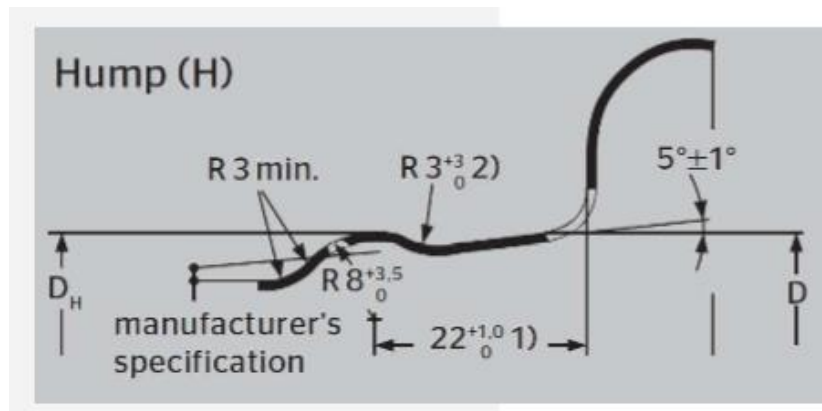


Figure 23

- Silicon main alloy element
- Excellent fluidity and casting properties

5.1.2 Connection between damper and chassis

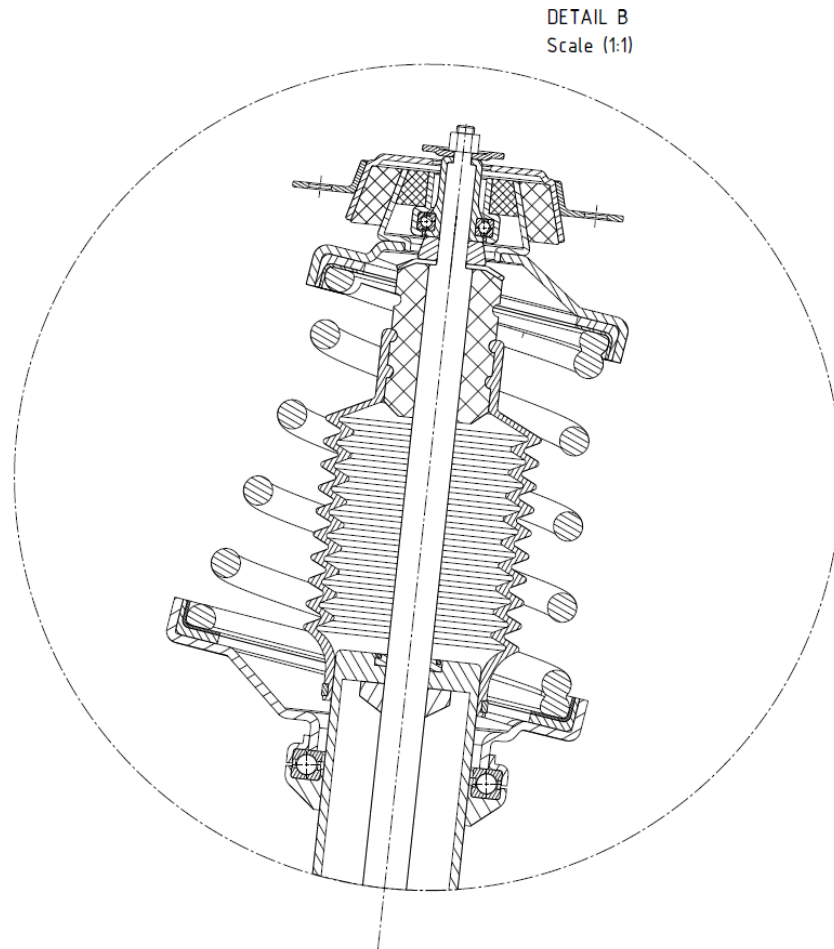


Figure 26

- The coil spring sits between two metal plates; at the two ends, between these two elements, an additional rubber layer and thin metal plate can be seen
- A deep groove radial bearing is observed in the upper part of the damper, it sustains radial loads while the polymeric/rubber material acts as a spherical joint
- A thrust bearing can be observed just below the bottom plate holding the spring. It sustains high axial loads
- The two bearings and the polymeric material allow the damper to rotate as the wheel is steered

5.1.3 Hub + hub carrier

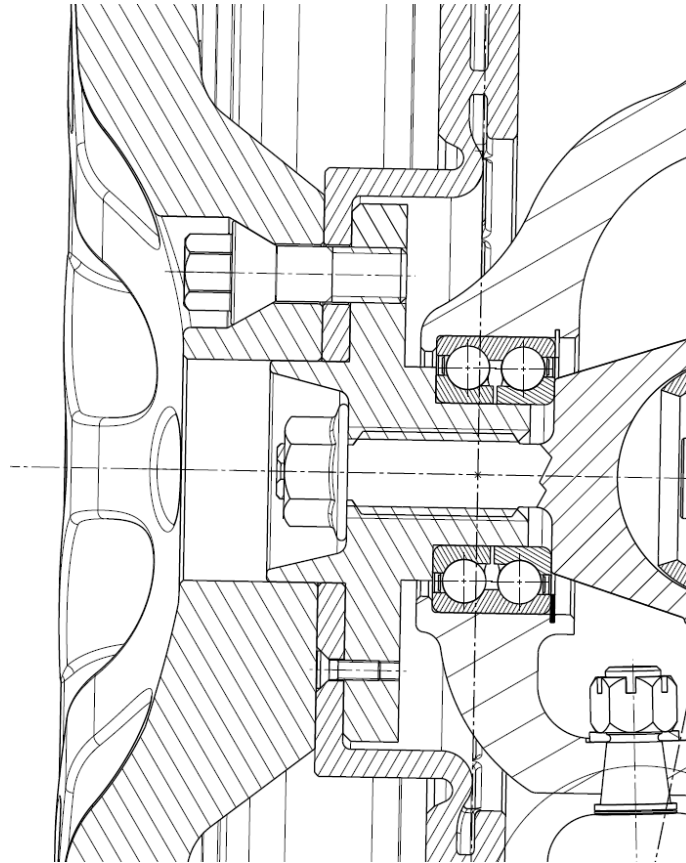


Figure 27

- Cast iron hub carrier
- Steel wheel hub manufactured with forging and eventually turning
- The bearing has O arrangement
 - o Preload realized through the nut
 - o External ring fixed by means of Sieger ring on the right side and by means of a shoulder on the left side

5.1.4 Constant velocity joint

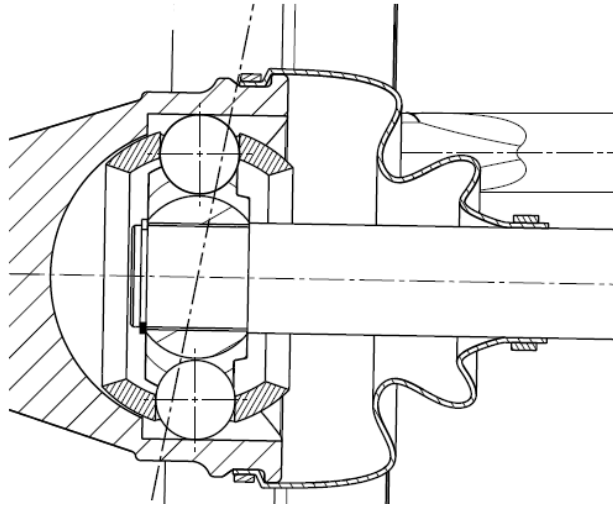


Figure 28

- The inner race is splined on the input transmission shaft and its axial displacement is prevented by the Segur ring. The balls are free to slide in axial direction along the outer race
- Torque flow: input shaft → inner race (through spline connection) → outer race (through steel balls) → wheel hub

6. Conclusion

The design and structural validation of the McPherson suspension system were successfully completed, meeting all kinematic and safety targets. Through a neural network-based optimization process, the suspension geometry was tuned to precisely match the required camber and steering angle curves. The lower control arm, designed for manufacturing via forging using Al 6061 alloy, underwent topological optimization to maximize stiffness while minimizing weight. Final FEM analysis confirmed the component's structural integrity, revealing a maximum Von Mises stress of 160.7 MPa and a vertical deflection of 2.3 mm, both well within the material's yield limits. The report concludes with a complete set of production drawings and tolerance specifications to ensure proper assembly and manufacturability.